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MYSQL

SUBSTRING(string, start, length)
or
SUBSTR(string, start, length)

Example

```
SELECT SUBSTRING('Database',1,4);
```

Result

Data

If omitted (MySQL)

```
SELECT SUBSTRING('Programming',5);
```

Result

ramming

Extracts everything from position 5 onward.

Negative Positions (Vendor Support)

Very useful interview topic.

MySQL

Supports

```
SELECT SUBSTRING('Database', -4);
```

Result

base

Counts from the end.

PostgreSQL

Negative positions are not supported.

SQL Server

Not supported.

SUBSTRING vs LEFT vs RIGHT

Function	Purpose
SUBSTRING	Any part
LEFT	Beginning only
RIGHT	End only

POSITION()

POSITION() finds where one string appears inside another.

Example

aadya@gmail.com

Find

@

Result

6

Think of it as SQL's indexOf().

```
SELECT POSITION('@' IN 'aadya@gmail.com');
```

Result

6

Unlike REPLACE(), which changes text, SUBSTRING() returns only a selected part.

MySQL Specifics

MySQL doesn't support POSITION() as the primary function in tutorials.

Instead you'll usually see

LOCATE ()

```
SELECT LOCATE('@', 'aadya@gmail.com');
```

Result

6

Word Search

```
SELECT POSITION('World' IN 'Hello World');
```



7

If not found



0

(MySQL)

POSITION + SUBSTRING

Very common.

Extract username

aadya@gmail.com

MySQL

```
SELECT SUBSTRING(  
Email,1,  
LOCATE('@',Email)-1  
)FROM Users;
```

CONCAT()

11.1 Introduction

CONCAT() joins multiple strings into one.

Example

Aadya	Raj
-------	-----



AadyaRaj

Usually with spaces



Aadya Raj

CONCAT vs ||

ANSI SQL

```
FirstName || ' ' || LastName
```

MySQL

Use

CONCAT ()

CONCAT_WS()

Means

Concatenate With Separator

Instead of

```
CONCAT(  
  FirstName, ' ',  
  LastName  
)
```

write

```
CONCAT_WS(' ',  
  FirstName,  
  LastName  
)
```

Output

Aadya Raj

```
CONCAT_WS(' ',  
  City, State,  
  Country  
)
```



Jaipur, Rajasthan, India

NULL Handling

Very important.

Suppose

FirstName	Aadya
LastName	NULL

MySQL

```
SELECT CONCAT(  
  FirstName, ' ',  
  LastName  
);
```



NULL

because any NULL argument makes the result NULL.

CONCAT_WS() behaves differently.

It skips NULL values.

```
CONCAT_WS(' ', 'Aadya', NULL, 'Raj'  
)
```



Aadya Raj